

Ligjerata 5a

Sistemi inercial

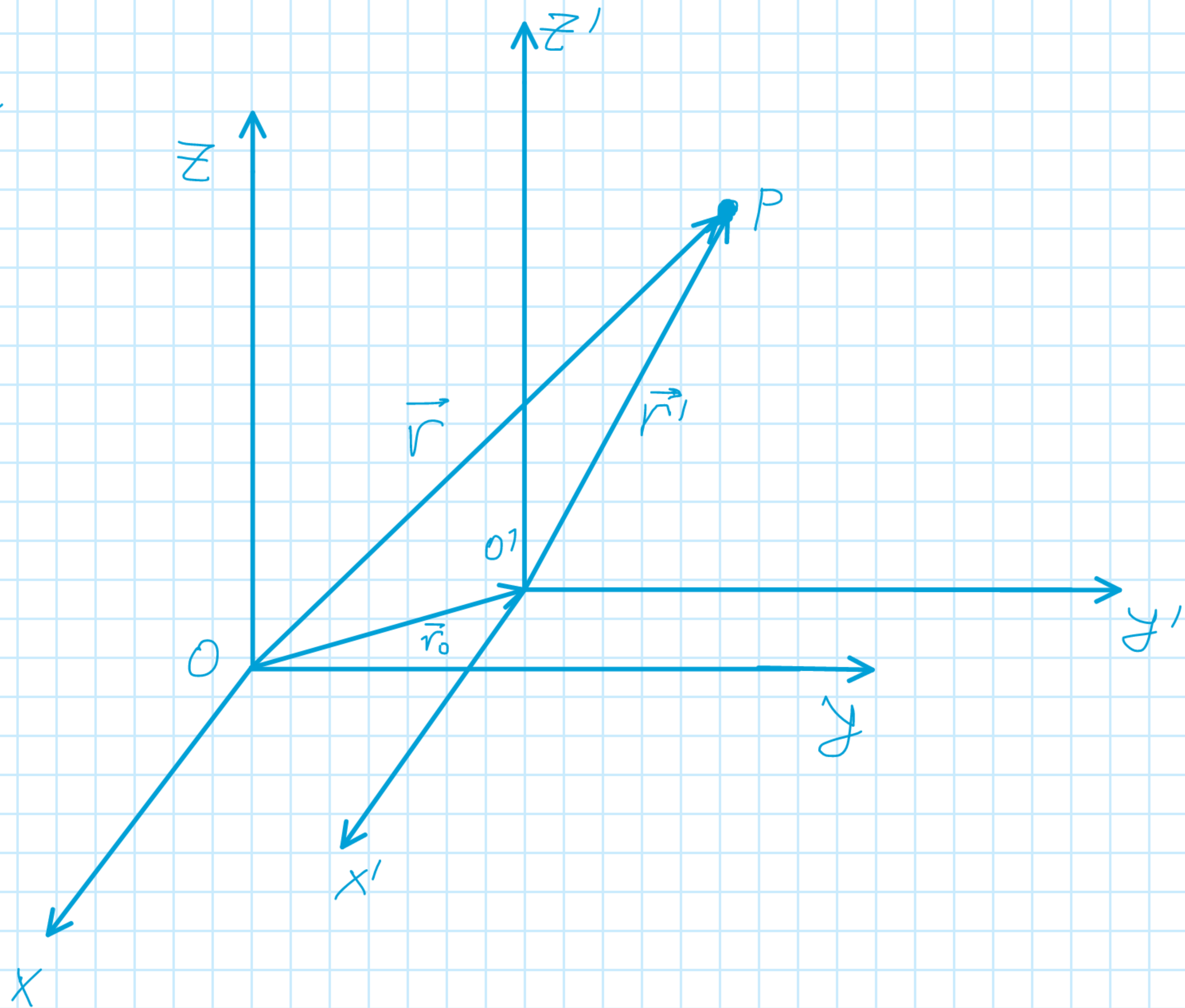
x, y, z sistemi koordinativ në përdhë

x', y', z' ————— në lëvizje ndaj
sistemit të parë $\vec{U} = \dot{\vec{r}}_0$

$$\vec{F} = m \ddot{\vec{r}}$$

$$\vec{r} = \vec{r}_0 + \vec{r}_1$$

a) $\vec{U} = \text{konst}, \dot{\vec{U}} = 0, \ddot{\vec{r}}_0 = 0$
 $\ddot{\vec{r}} = \ddot{\vec{r}}_1, \vec{F} = \vec{F}'$



b) $\vec{U} \neq \text{konst} \quad \dot{\vec{U}} = \vec{a}_0 = \ddot{\vec{r}}_0$

$$\vec{F} = m \ddot{\vec{r}} = m (\ddot{\vec{r}}_0 + \ddot{\vec{r}}_1) = m \ddot{\vec{r}}_0 + m \ddot{\vec{r}}_1$$

$$\vec{F} = \vec{F}_0 + \vec{F}', \quad \vec{F}_0 = -m \vec{a}_0$$

$$\vec{F}' = \vec{F} - \vec{F}_0$$

$$\vec{a} = \frac{d\vec{U}}{dt} = \dot{\vec{U}}$$

$$\vec{a} = \frac{d}{dt} \left(\frac{d\vec{r}}{dt} \right)$$

$$\vec{a} = \frac{d^2 \vec{r}}{dt^2} = \ddot{\vec{r}}$$

